

8 Chapters · 160+ Questions · 3 Quiz Levels · Free Access · No Account Required

COURSE OVERVIEW — CHAPTER BY CHAPTER

1. **RTAC Architecture** — The SEL RTAC (SEL-3505, SEL-3530, SEL-3555) is a protocol gateway, data concentrator, and local logic engine in one ruggedized DIN-rail device. This chapter covers hardware variants, I/O modules, communication ports, and the commissioning sequence before any software is opened.
2. **ACSELERATOR Architect** — SEL's configuration software organizes every RTAC project as a database: tags, aliases, point mapping, and revision control. Engineers learn the project structure that every RTAC configuration follows and the workflow that prevents configuration drift.
3. **DNP3 Configuration** — The RTAC communicates upstream to the SCADA master via DNP3. This chapter covers master and outstation roles, point database mapping, event handling, unsolicited reporting, and the DNP3 parameters that determine how quickly a SCADA system sees a field change.
4. **Modbus Gateway** — The RTAC translates Modbus TCP requests into DNP3 point values. This chapter covers server configuration, register mapping from DNP3 to coils and holding registers, and the common integration patterns for legacy SCADA systems that speak only Modbus.
5. **IEC 61850 Integration** — GOOSE messages carry protection relay trip signals across the station LAN in microseconds. Sampled Values carry current and voltage waveforms. MMS handles operator control. This chapter covers all three and how RTAC maps IEC 61850 data to its internal tag database.
6. **Local Logic** — When communications to the SCADA master fail, the RTAC continues executing local IEC 61131-3 programs. This chapter covers the ACSELERATOR logic engine: event-driven automation, failover sequences, and the programs that keep a substation safe when the network is down.
7. **Security & Access Control** — Role-based access, RBAC profile configuration, serial port security, and firmware update procedures. The RTAC is a critical infrastructure device; this chapter covers the security posture required for NERC CIP compliance in power utility applications.
8. **Field Scenarios** — Three substation automation case studies: transformer bay control with automatic tap changer response, feeder protection coordination with downstream reclosers, and load shedding automation that executes correctly under a communication fault to the SCADA master.

Certification Relevance

- **SEL Certified Engineer** — SEL offers application engineer certification through their training center; RTAC configuration is a core track.
- **GICSP (GIAC)** — Substation automation, IEC 61850 security, and protocol gateway configuration appear in ICS security certification exam domains.
- **ISA CCST** — Protocol gateway integration and field device configuration are tested topics at CCST Levels II and III.

Next Steps

1. Complete the DNP3 and IEC 61131-3 guides alongside RTAC — both protocol knowledge and programming skills are prerequisites for field configuration.
2. Download ACSELERATOR Architect (free from selinc.com) and configure a lab project against a virtual RTAC.
3. Attend SEL application engineering training (Pullman, WA or regional); the in-person lab is the fastest path to RTAC proficiency.
4. Pursue GICSP for the ICS security credential most relevant to power utility and substation work.